Week = 2

06/10/26

1. Implement Tic-Tac-Toe game in which computer uses MINIMAX algorithm to select the best move

```
TIC-TAC-TOE (state)
    if TERMINAL-TEST (state) then
        return UTILITY (state)
    if PLAYER (state) = MAX then
        return MAX-VALUE (state)
    else
        return MIN-VALUE (state)

MAX-VALUE (state)
    if TERMINAL-TEST (state) then
        return UTILITY (state)
    v ← -∞
    for each action in ACTIONS(state) do
        v ← MAX (v, MIN-VALUE(RESULT (state, action)))
    return v

MIN-VALUE (state)
    if TERMINAL-TEST (state) then
        return UTILITY (state)
    v ← +∞
    for each actions in ACTIONS (state) do
        v ← MIN(v, MAX-VALUE (RESULT (state, action)))
    return v
```

MINIMAX - DECISION (state)

Return the action a in ACTIONS (state) with maximum MIN-VALUE (RESULT (state, a))

UTILITY (state)

if MAX has won then
    return +1
if MIN has won then
    return -1
return 0

Test cases :-

1. Computer winning :

Input	Output
O O _	O O O
X X _	X X _
_ _ _	_ _ _

Computer wins

2. Computer must block :

Input	Output
X X _	X X O
O _ _	O _ _
_ _ _	_ _ _

3. Draw :

Input	Output
X O X	X O X
X O O	X O O
O X _	O X O

Game Draw

6.10.20